

How Man in the Middle Attacks Cause Damage

Security breaches in a computer network can have disastrous results. They may be the cause of company secrets leaking out or result in industrial espionage. A common breach is the Man in the Middle or MITM attack. This happens when malicious players insert themselves in a network and intercept all incoming and outgoing traffic, either passively or actively.



An MITM attack is when a malicious user eavesdrops on the conversation or modifies the conversation between two parties interacting in cyber space. When the malicious user is only eavesdropping on the conversation, it is a passive attack, but modifying the conversation is considered an active attack. Consider two users Alice, Bob and a malicious attacker. If an MITM attack is launched in this case, any conversation between Alice and Bob will go through the attacker.

In this case, the attacker is just looking at the conversation. Therefore, this is a passive attack, which is difficult to detect as no data is modified.

This attack can also be extended between a Web server and a client. Consider a scenario in which Alice is looking forward to transferring US\$ 1000 to Bob online. A malicious user named James can launch an MITM attack and manipulate the conversation between the bank server and Alice to transfer the US\$ 1000 to himself (James) instead. As the data is being modified, this is an active attack.

When comparing active and

passive attacks, the latter may seem less dangerous but they are not. A passive attack might lead to the revelation of sensitive information to unauthorised (or unwanted) people. As the information is not tampered with, it is difficult to detect the information leakage. For example, consider the situation of a question paper leakage. Though no data in the question paper is tampered with, the leakage reduces

the credibility and quality of the exam. Therefore, a passive attack is as dangerous as an active attack.

This article describes how a passive MITM attack is launched and some of the security measures that can be taken against the attack. We have used the following open source tools to explain MITM: EtherApe, ARPspooof, ssl-strip and Driftnet.

Each tool is used for a particular purpose:

1. Network monitoring is done using EtherApe
2. An MITM attack is launched using ARPspooof
3. The victim is monitored using Driftnet

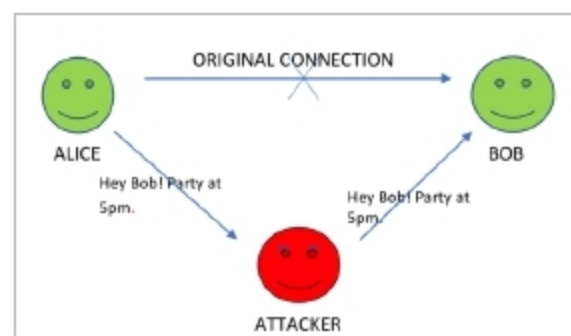


Figure 1: Passive attack

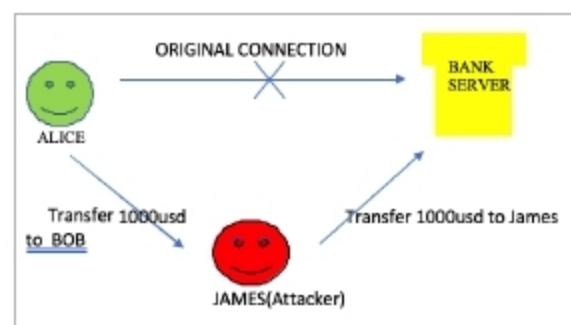


Figure 2: Active attack

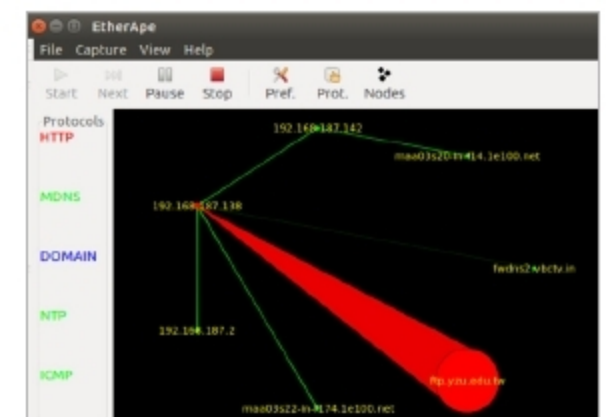


Figure 3: EtherApe